

LOS INTERCEPTOR GUIDANCE FOR DEFENDING AN AIRCRAFT

PRINCIPLES OF GUIDANCE FOR
AUTONOMOUS VEHICLES_PROJECT

Group number : 3

Agenda

Topics Covered

1. INTRODUCTION
2. 3-BODY GUIDANCE PROBLEM
3. LOS GUIDANCE-MOVING LAUNCH STATION
4. SPEED RELATIONS
5. LATERAL ACCELERATION
6. ADVANTAGE OF LOS
7. COMMAND LOS
8. CO-OPERATIVE GUIDANCE LAW
9. RESULTS

INTRODUCTION

- Aircraft face missile threats with high speed and maneuverability, making passive defenses (e.g., flares, chaff) limited.
- Aircraft can counter the missile threat by launching a defending missile, creating a three-body interception scenario (aircraft, attacker, defender).
- The main objective of the paper is to investigate Line-of-Sight (LOS) guidance as a strategy for a defending missile to intercept an incoming missile, with the defending missile launched from a moving aircraft.
- Another objective is to develop a cooperative guidance law that increases the difficulty for the attacking missile to evade interception by the defending missile.

3 Body Guidance Problem

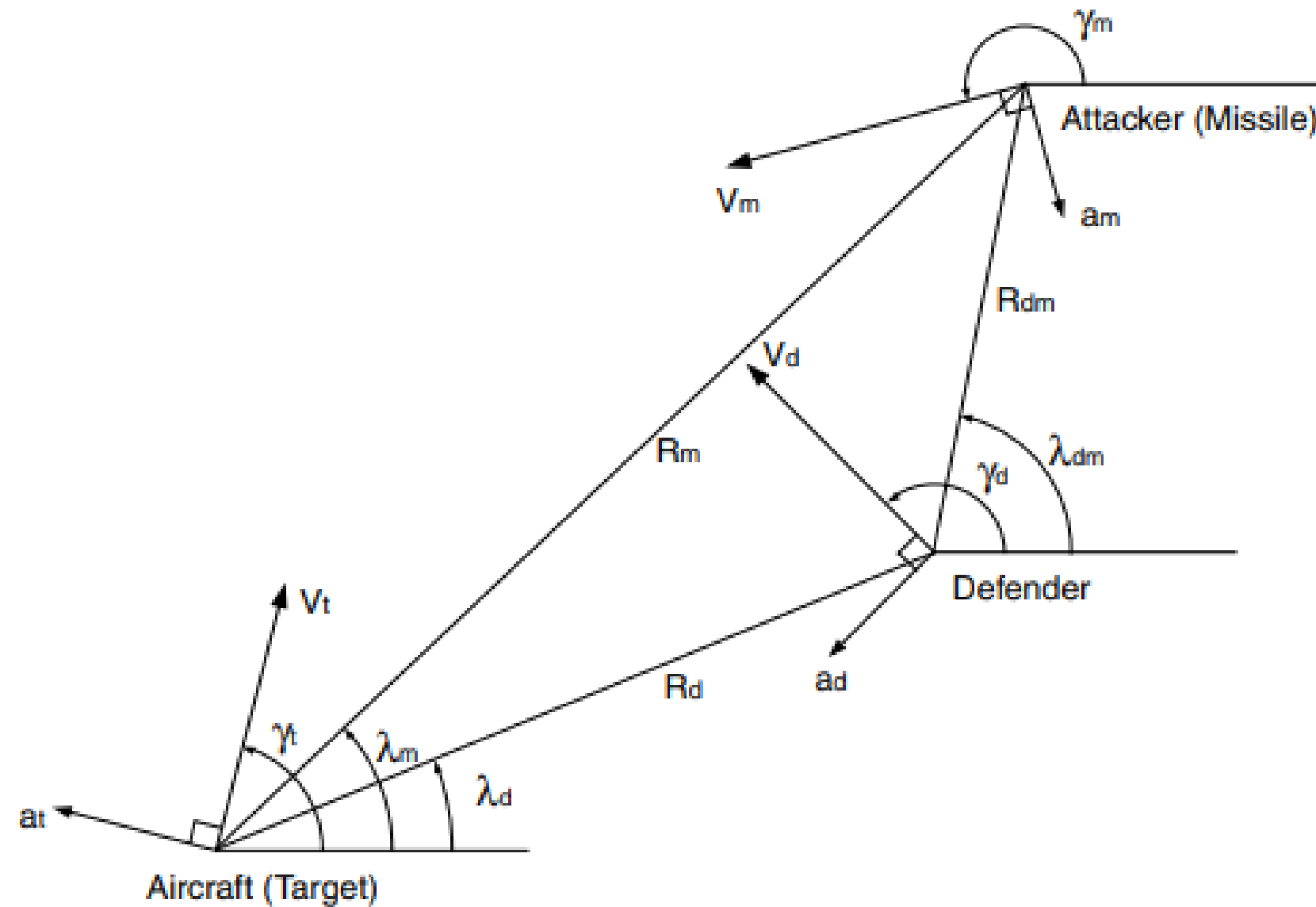


Fig. 1 Three-body engagement scenario.

$$\dot{\lambda}_m = \frac{v_m \sin(\gamma_m - \lambda_m) - v_t \sin(\gamma_t - \lambda_m)}{R_m}$$

$$\dot{\gamma}_m = \frac{a_m}{v_m}$$

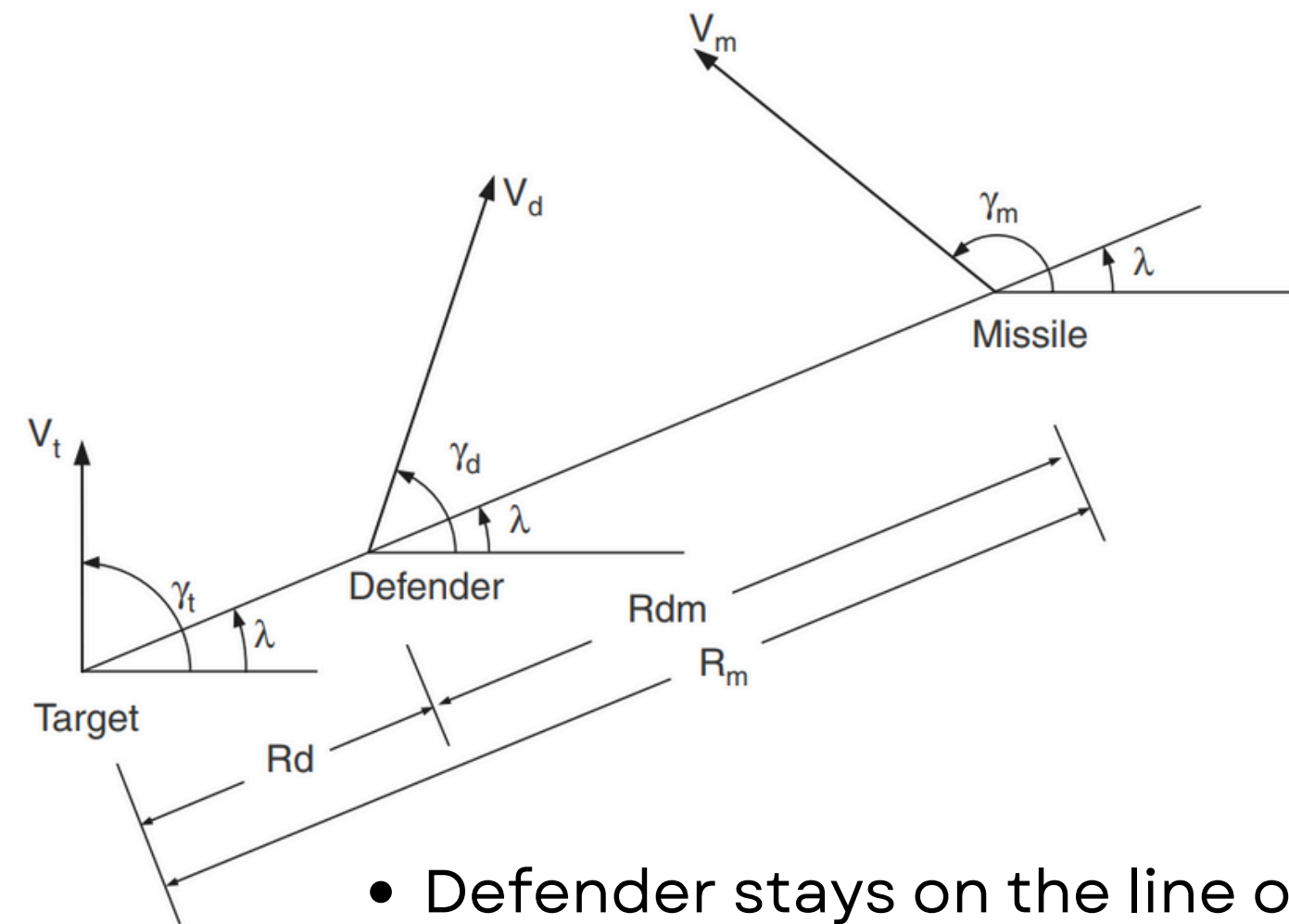
$$\dot{\lambda}_d = \frac{v_d \sin(\gamma_d - \lambda_d) - v_t \sin(\gamma_t - \lambda_d)}{R_d}$$

$$\dot{\gamma}_d = \frac{a_d}{v_d}$$

$$\dot{\lambda}_{dm} = \frac{v_m \sin(\gamma_m - \lambda_{dm}) - v_d \sin(\gamma_d - \lambda_{dm})}{R_{dm}}$$

$$\dot{\gamma}_t = \frac{a_t}{v_t}$$

LOS Guidance - Moving Launch station



- Defender stays on the line of sight between aircraft and missile.

- All the line of sight angles become equal

$$\lambda = \lambda_d = \lambda_m = \lambda_{dm}$$

Velocity components along LOS

$$v_{tlos} = v_t \cos(\gamma_t - \lambda)$$

$$v_{dlos} = v_d \cos(\gamma_d - \lambda)$$

$$v_{mlos} = v_m \cos(\gamma_m - \lambda)$$

Velocity components normal to LOS

$$v_{tplos} = v_t \sin(\gamma_t - \lambda)$$

$$v_{dplos} = v_d \sin(\gamma_d - \lambda)$$

$$v_{mplos} = v_m \sin(\gamma_m - \lambda)$$

LOS Guidance - Moving Launch station

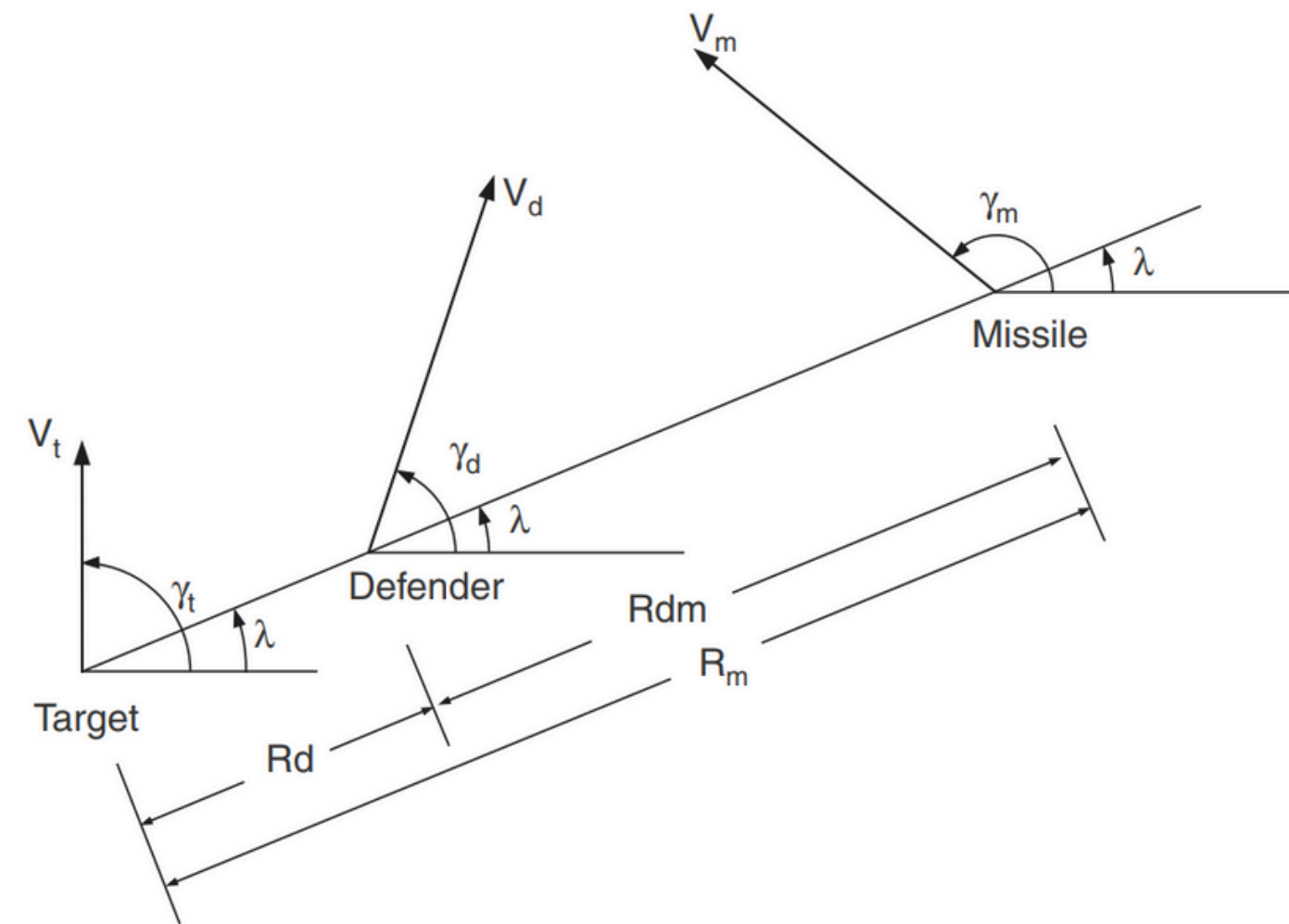


Fig. 2 LOS engagement geometry.

- The value of R_d starts from zero till it reaches R_m .

$$\frac{R_d}{R_m} \in [0, 1]$$

- LOS rate of rotation is expressed as :

$$\dot{\lambda} = \frac{v_{mplos} - v_{tplos}}{R_m} = \frac{v_{dplos} - v_{tplos}}{R_d} = \frac{v_{mplos} - v_{dplos}}{R_{dm}}$$

Speed Relations

- Velocity component of defender normal to LOS is solved from LOS rate of rotation expression.

$$v_{dplos}(t) = \frac{R_d}{R_m} v_{mplos}(t) + \left(1 - \frac{R_d}{R_m}\right) v_{tplos}(t)$$

- At launch, Defender velocity normal to LOS = Aircraft velocity normal to LOS.
- At interception, Defender velocity normal to LOS = Missile velocity normal to LOS.

Lateral Accelerations

- Differentiating angular velocity expressions of missile and defender with respect to time and solving gives the component of lateral acceleration of the defender normal to LOS.

$$a_{dplos} = \frac{R_d}{R_m} a_{mplos} + \left(1 - \frac{R_d}{R_m}\right) a_{tplos} + 2\dot{\lambda} \left(\dot{R}_d - \frac{R_d}{R_m} \dot{R}_m \right)$$

- This acceleration consists:
 - Weighted sum of lateral accelerations of missile and target normal to LOS
 - A term proportional to LOS rate.
- Note: This acceleration is derived assuming defender is already on the line of sight joining missile and target.

Strategy Advantage

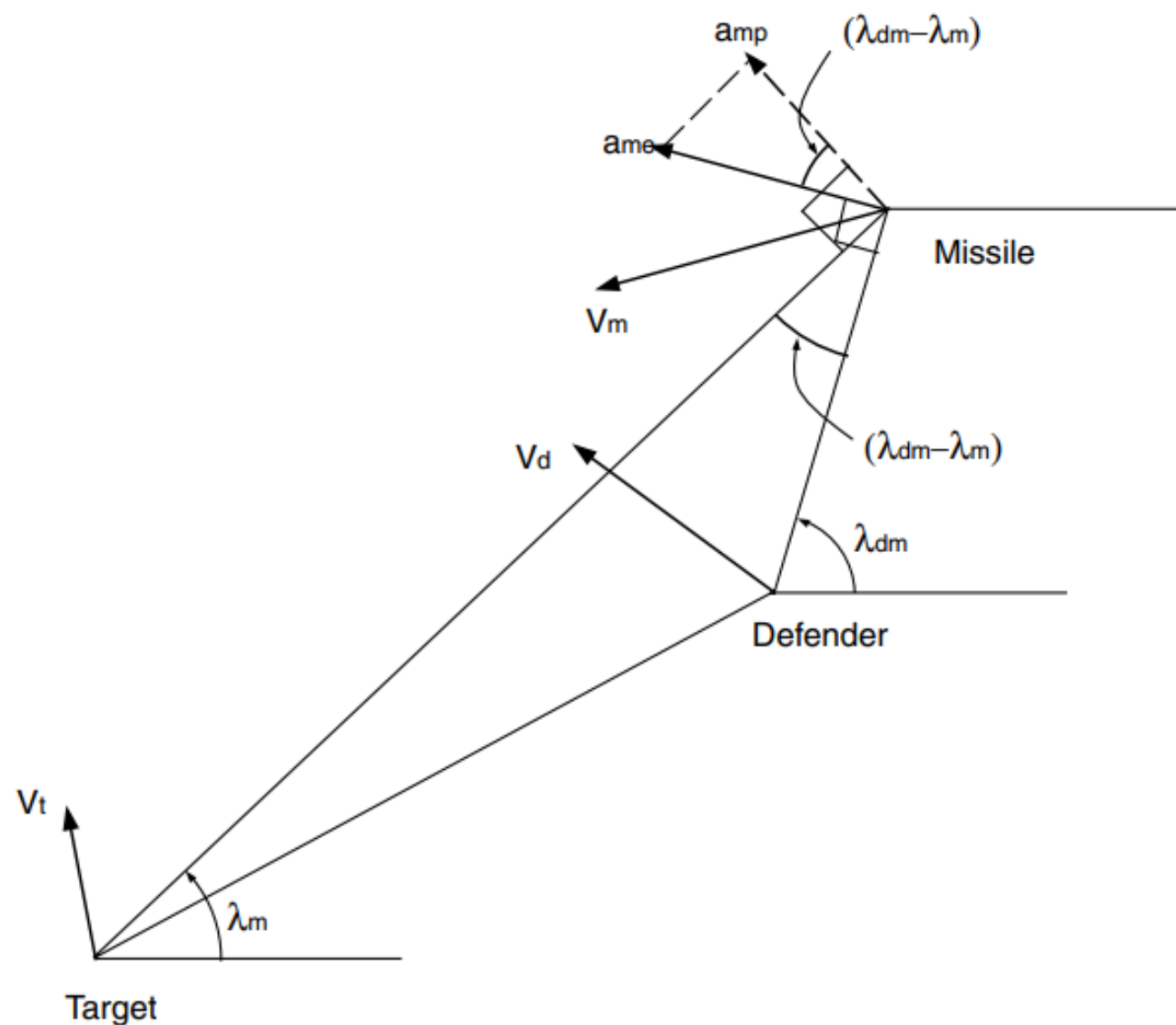
- Imagine missile performs an evasive measure to escape the defender with an acceleration a_{me} normal to LOS.
- Distance traveled due to this maneuver R_p in time Δt :

$$R_p = \frac{a_{me}}{2} \Delta t^2 \cos(\lambda_{dm} - \lambda_m)$$

- R_p is maximum when $\lambda_{dm} = \lambda_m$
- The penalty for an evasive maneuver to be performed by the missile is maximum when the LOS angles are equal, that is when defender follows **Line of Sight Guidance**.

Command LOS

- In real cases, LOS geometry needs to be achieved by other methods due to delays and estimation errors
- Command LOS is implemented to overcome this problem - a component proportional to the difference needed for the defender to cover in order to achieve LOS geometry is added.



$$a_d = KG(s)R_d(\lambda_m - \lambda_d) + a_t + R_d\ddot{\lambda}_m + 2\dot{R}_d\dot{\lambda}_m$$

Where,

$$G(s) = \frac{1 + (s/a)}{1 + (s/b)}$$

Command LOS

- This acceleration obtained from CLOS can be implemented in an actual control system as follows:

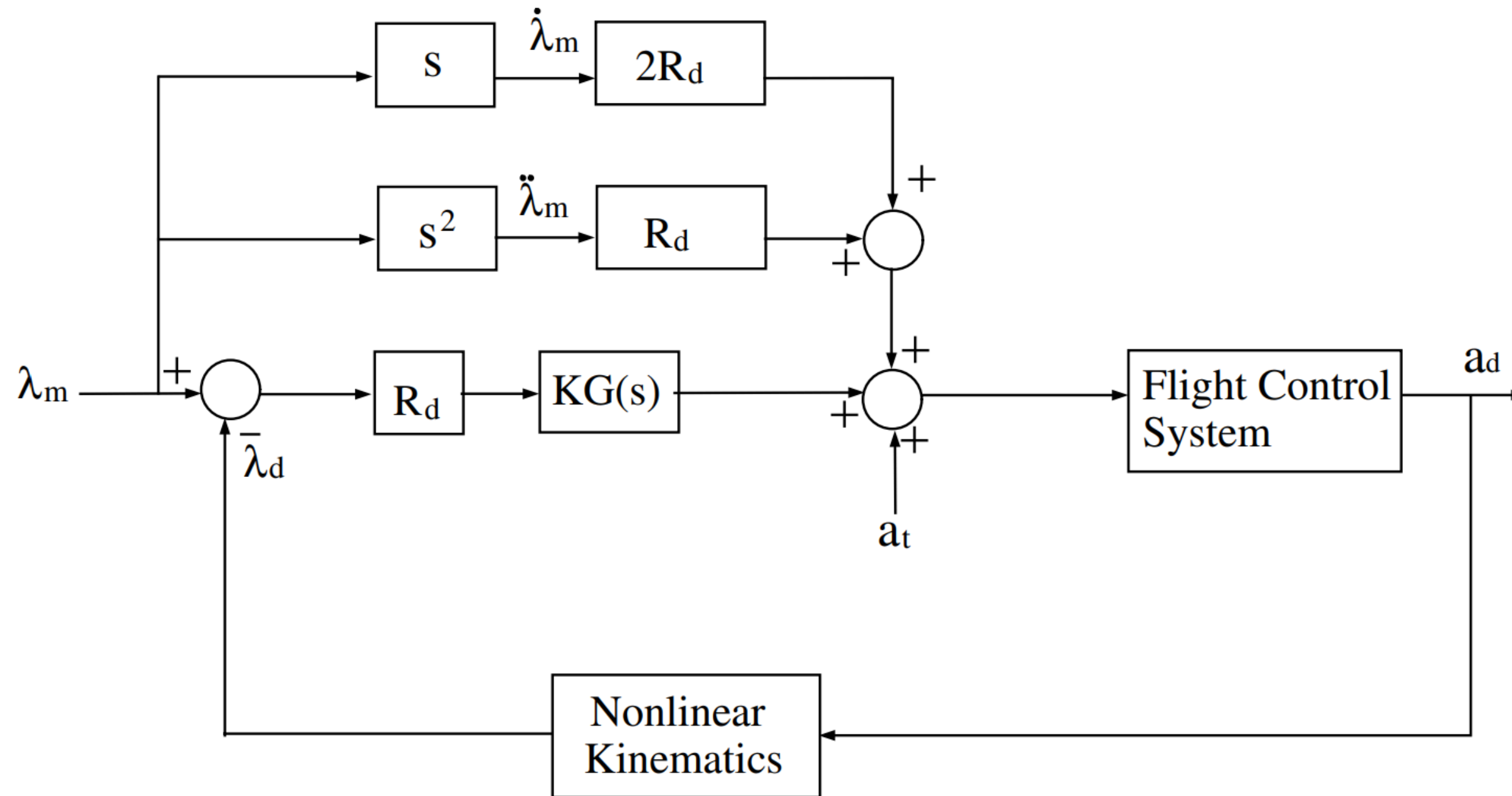


Fig. 4 Block diagram for defender CLOS implementation.

Cooperative Guidance Law

- Let the aircraft use PN like guidance law to avoid the missile which is using PN to pursue the aircraft.

$$a_{\text{tplos}} \simeq a_t = -\mu \dot{\lambda} \quad \text{and} \quad a_{\text{mplos}} = \eta \dot{\lambda} \cos(\gamma_m - \lambda) \simeq -\eta \dot{\lambda}$$

- Objective is to maximize missile lateral acceleration and hence maximize μ and also maintain ratio of defender to missile acceleration to be less than 1.
- This ratio is obtained from the equation

$$\frac{a_{\text{dplos}}}{a_{\text{mplos}}} = \frac{R_d}{R_m} + \frac{\mu(1 - R_d/R_m) - 2[\dot{R}_d - (R_d/R_m)\dot{R}_m]}{\eta}$$

- Maximum μ such that ratio is < 1 is to be found. The target acceleration for such value is found to be

$$a_t = -2\dot{R}_d \dot{\lambda}$$

Simulations

- Planar engagement scenario with constant speed point mass vehicles.
- The equations of motion are given as

$$\dot{x}_i = v_i \cos \gamma_i \quad \dot{y}_i = v_i \sin \gamma_i \quad \dot{\gamma}_i = \frac{a_i}{v_i}$$

where i represents t , m and d for target, missile and defender respectively.

- All the initial velocities and separations:

$$v_d = v_m = 600 \text{ m/s} \quad R_0 = 5000m$$

$$v_t = 200 \text{ m/s}$$

- All 3 bodies will be launched at the same time $t = 0$
- Aircraft and defender start from the same position.
- CLOS is implemented but without the $G(s)$ term, hence K is to be tuned for every simulation separately.

Simulations

Three standard attack geometries are considered

- Side on, Head on and Tail chase

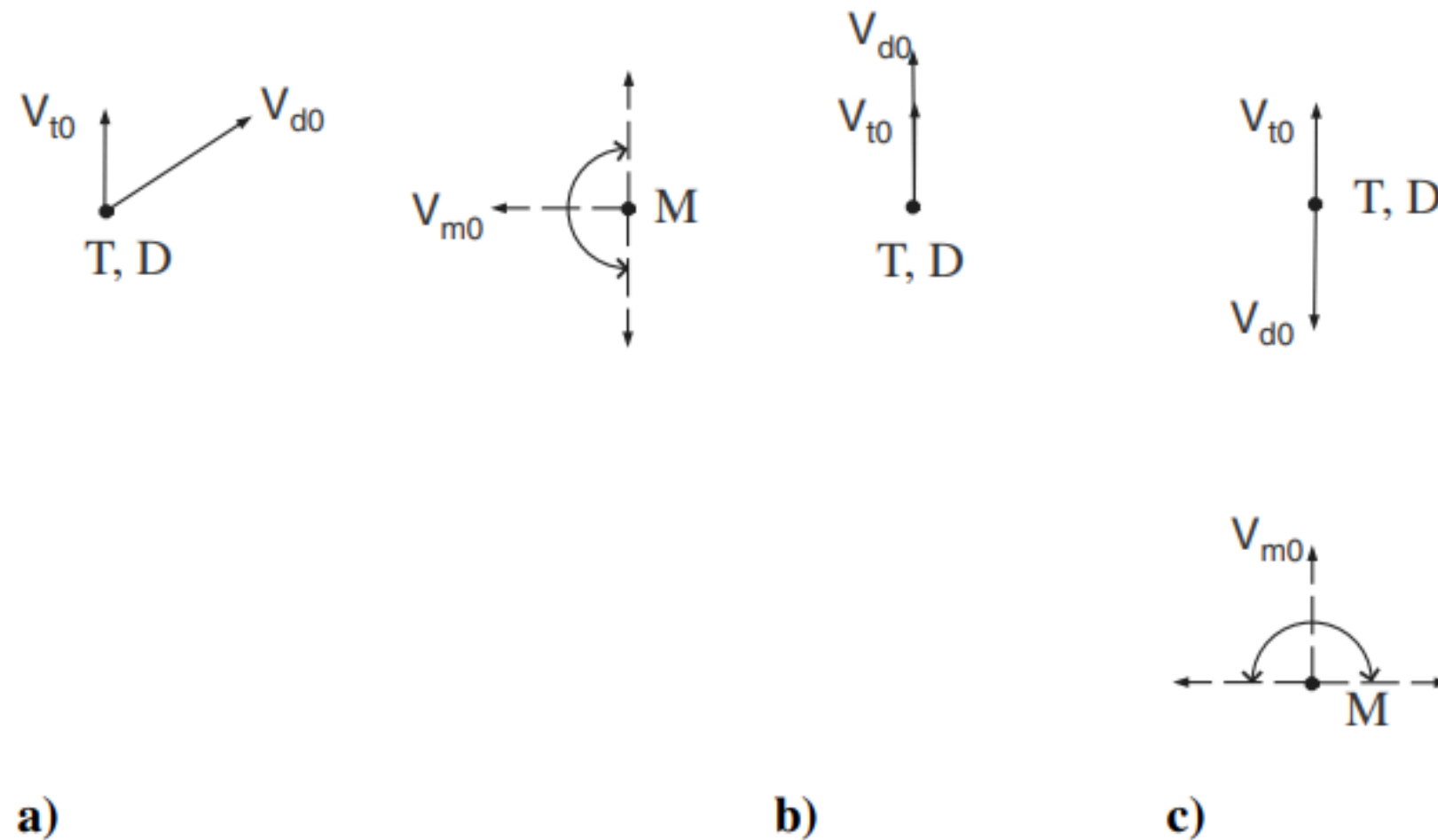


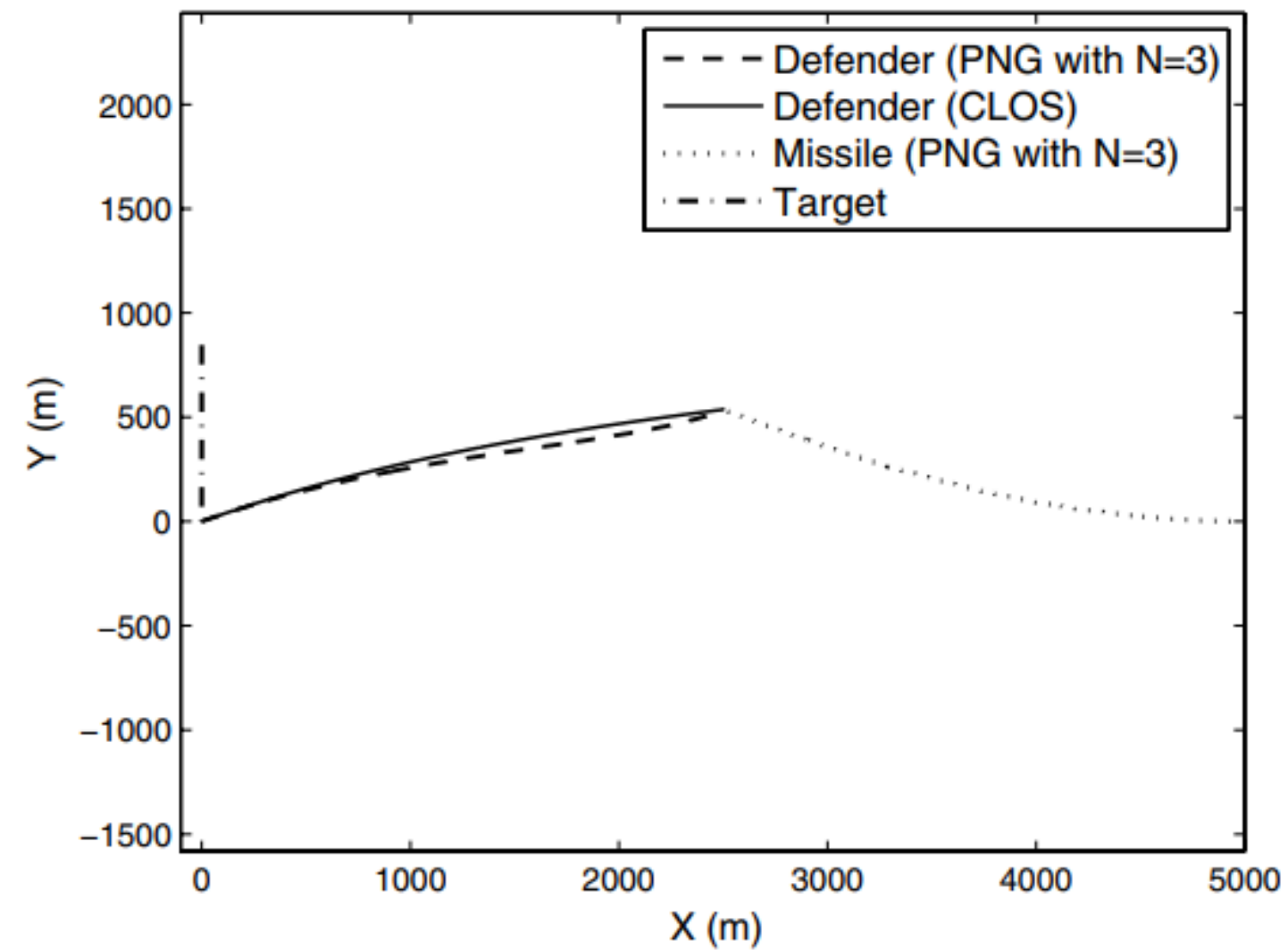
Fig. 5 Attack geometries: a) side on, b) head on, and c) tail chase.

Table 1 Engagement parameters for the three attack geometries

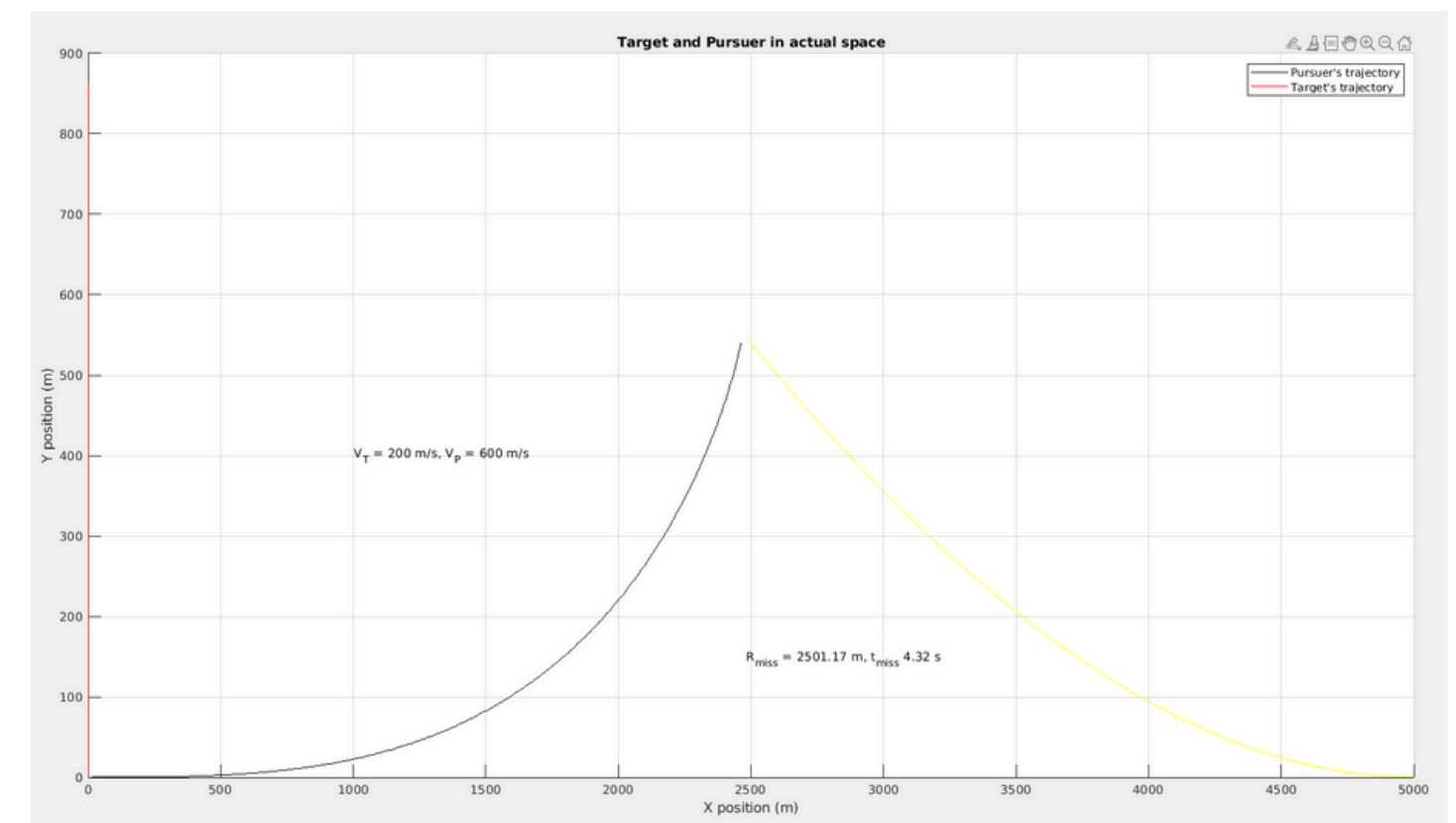
Parameter	Side-on geometry	Head-on geometry	Tail-chase geometry
λ_{t0} , deg	0	90	270
$R_d(0) = R_{dm}(0)$, km	5	5	5
γ_{t0} , deg	90	90	90
γ_{d0} , deg	19.47	90	270
$\gamma_{m0} \in$, deg	(90, 270)	(180, 360)	(0, 180)

Results

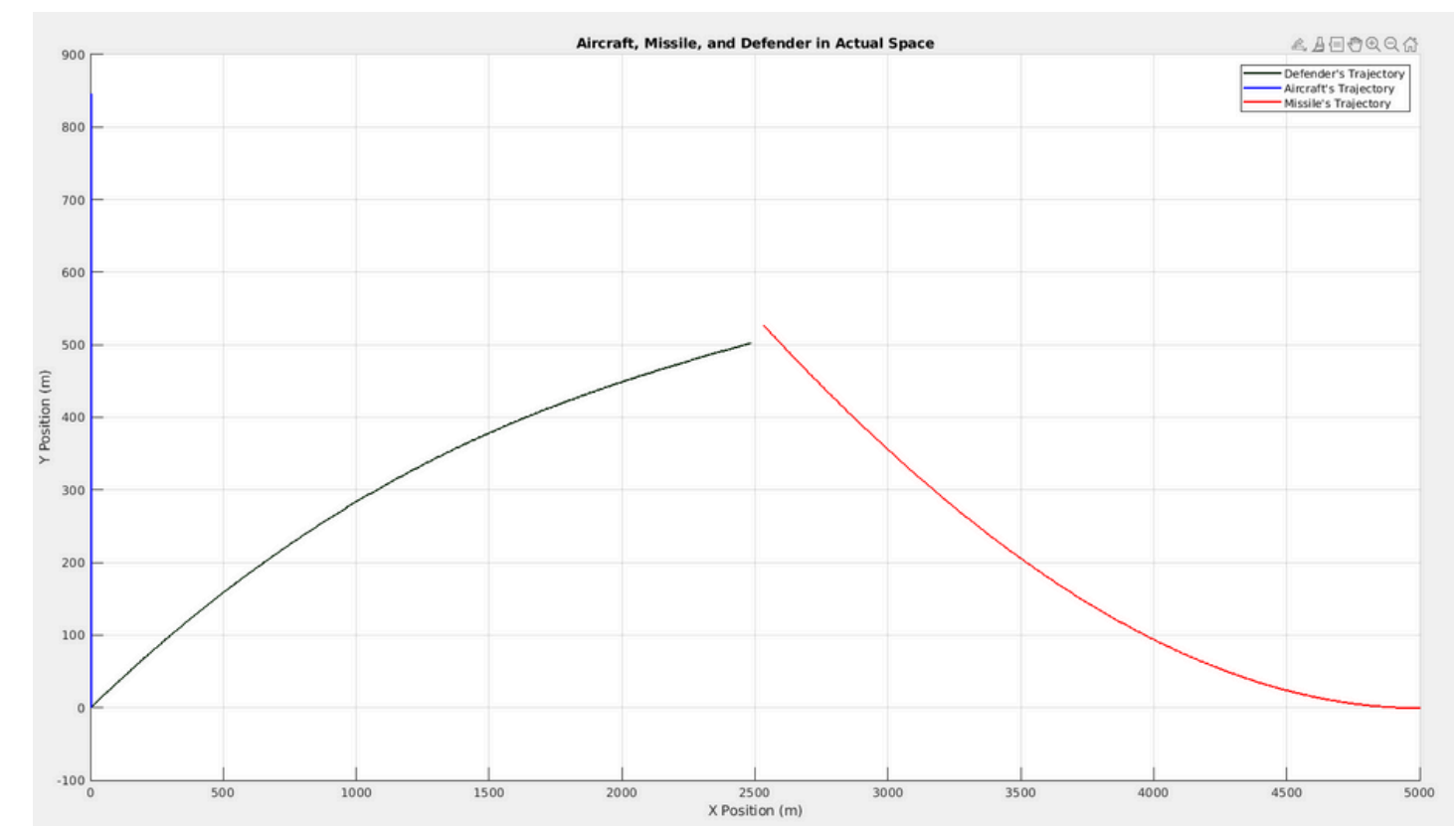
Side-on attack geometry



a) Trajectories



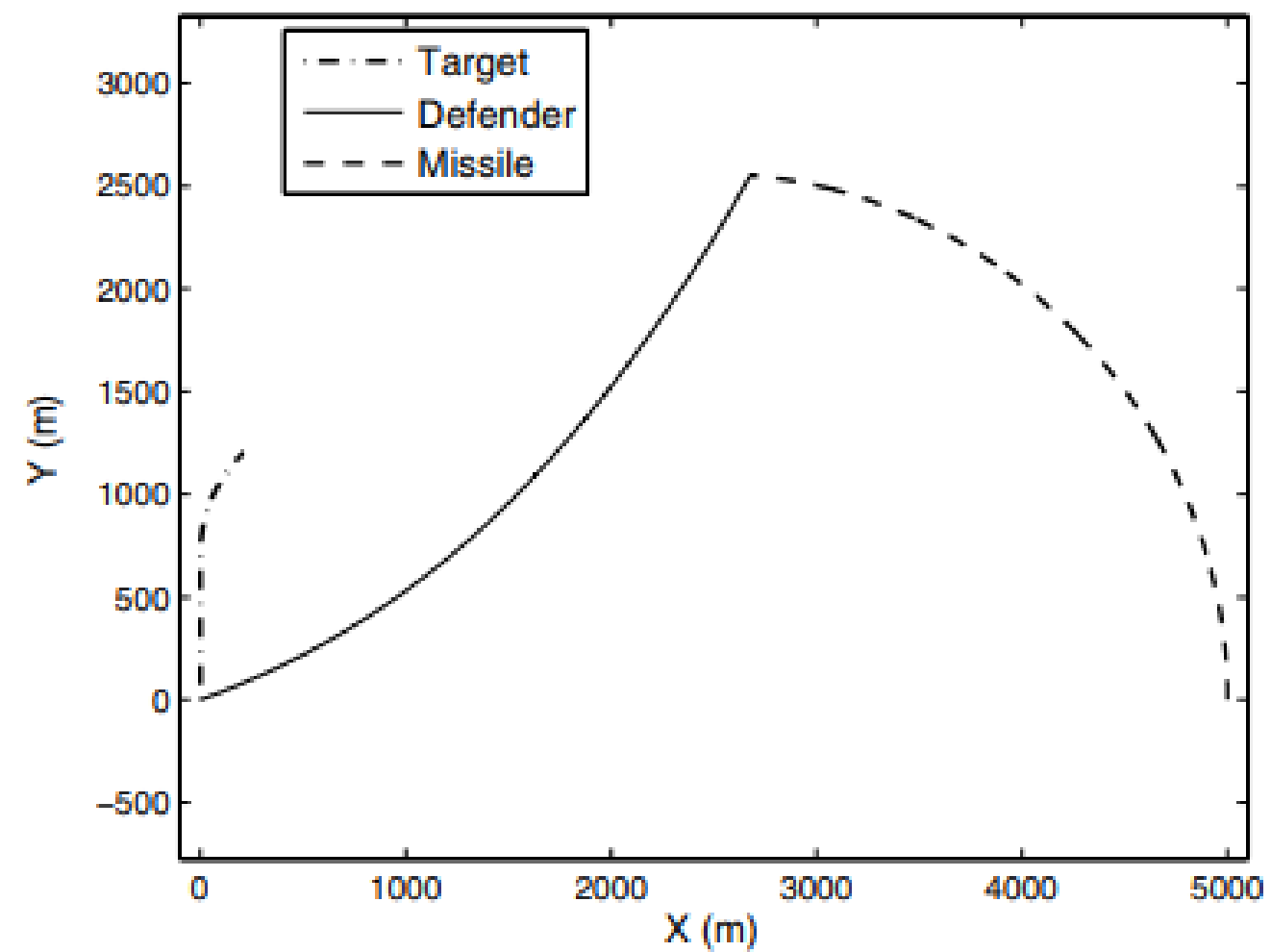
Trajectory with PN Geometry



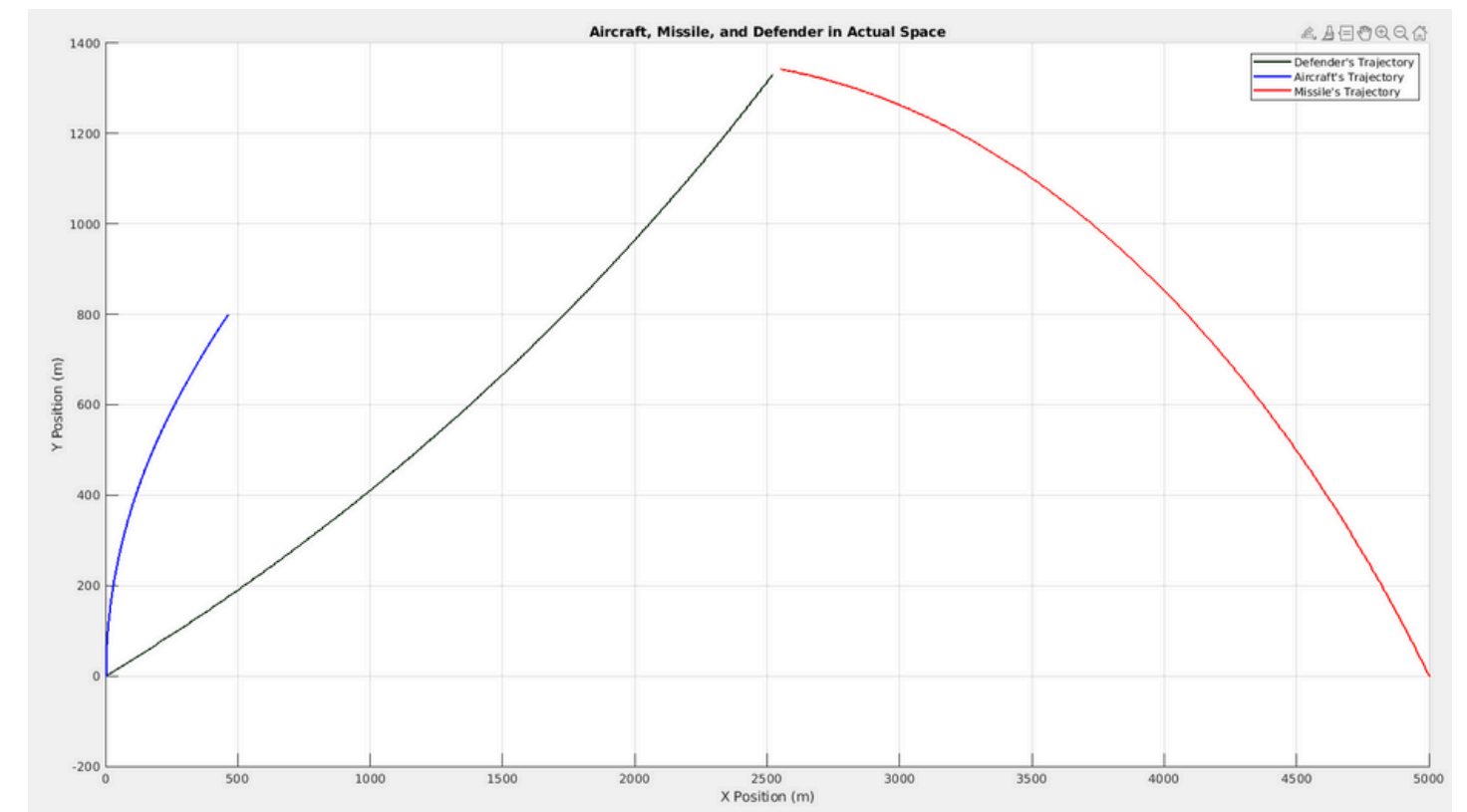
Trajectory with LOS Geometry

Results

Side-on attack geometry



a) Trajectories



Trajectory with CLOS Geometry

Results

Side-on attack geometry

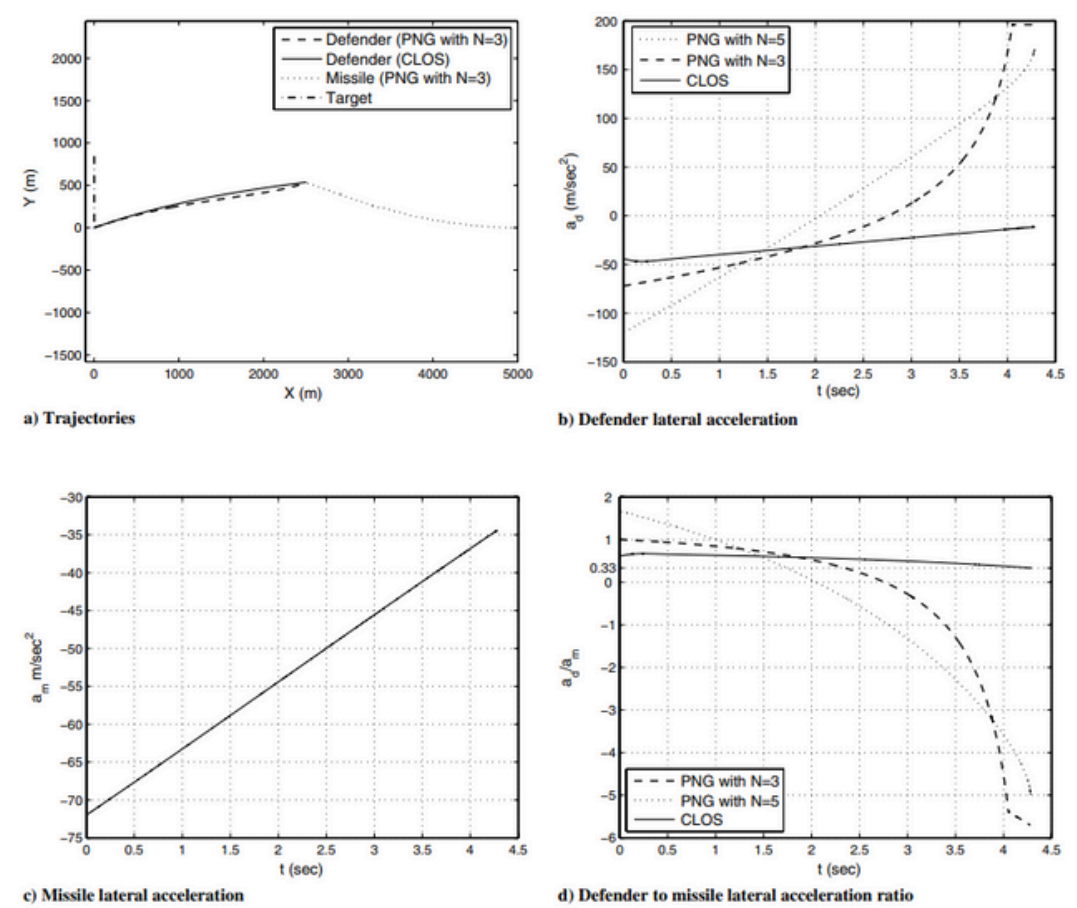
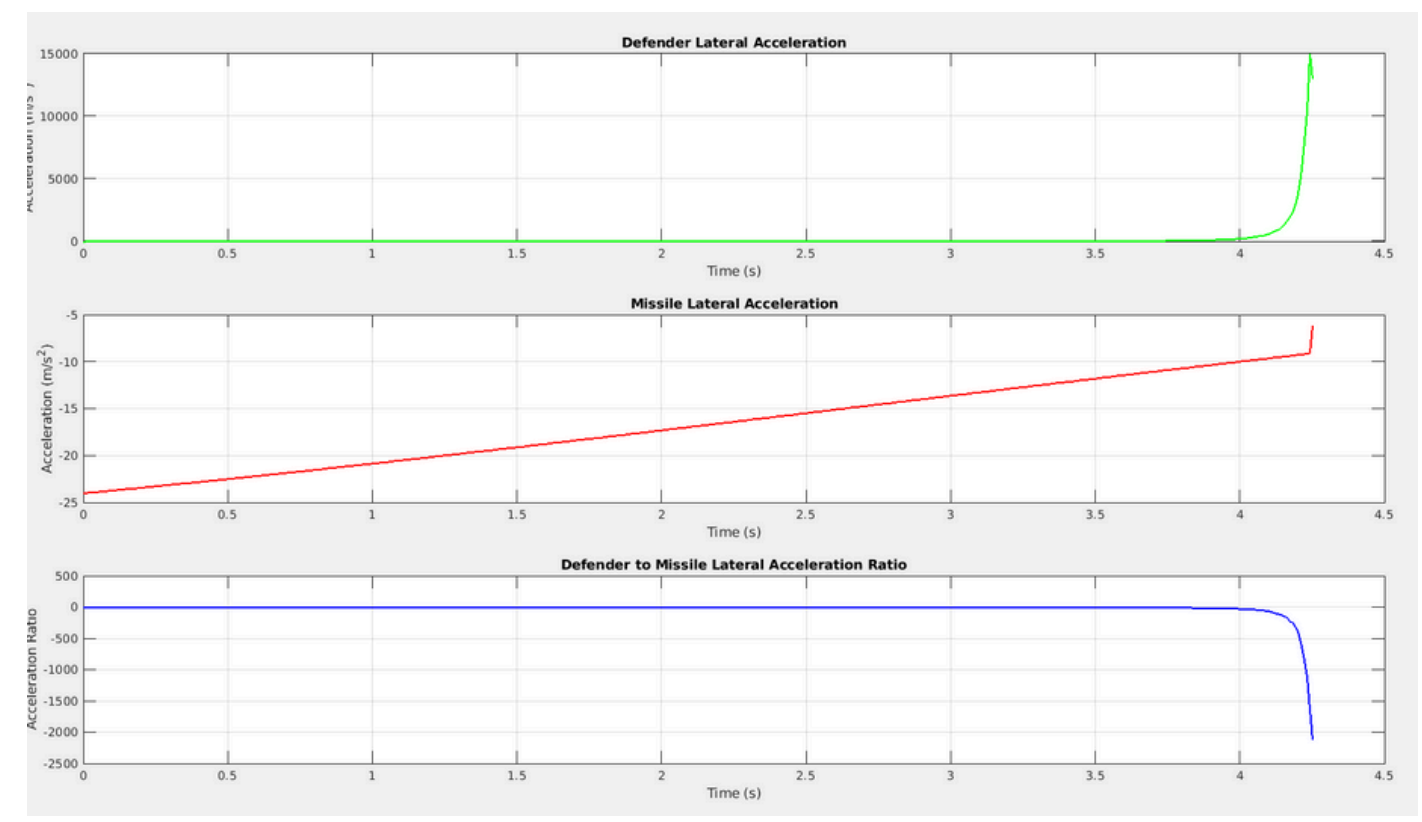
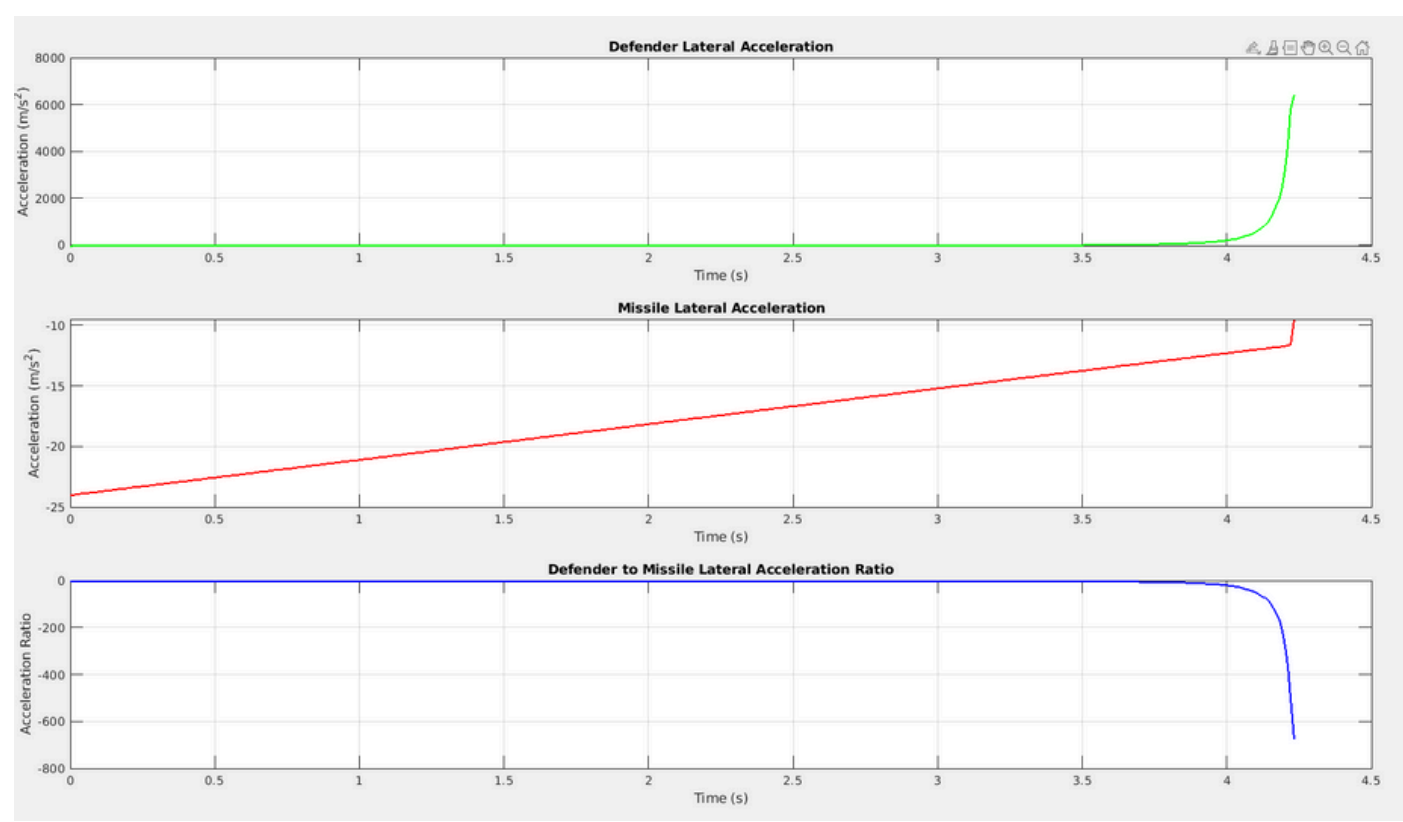


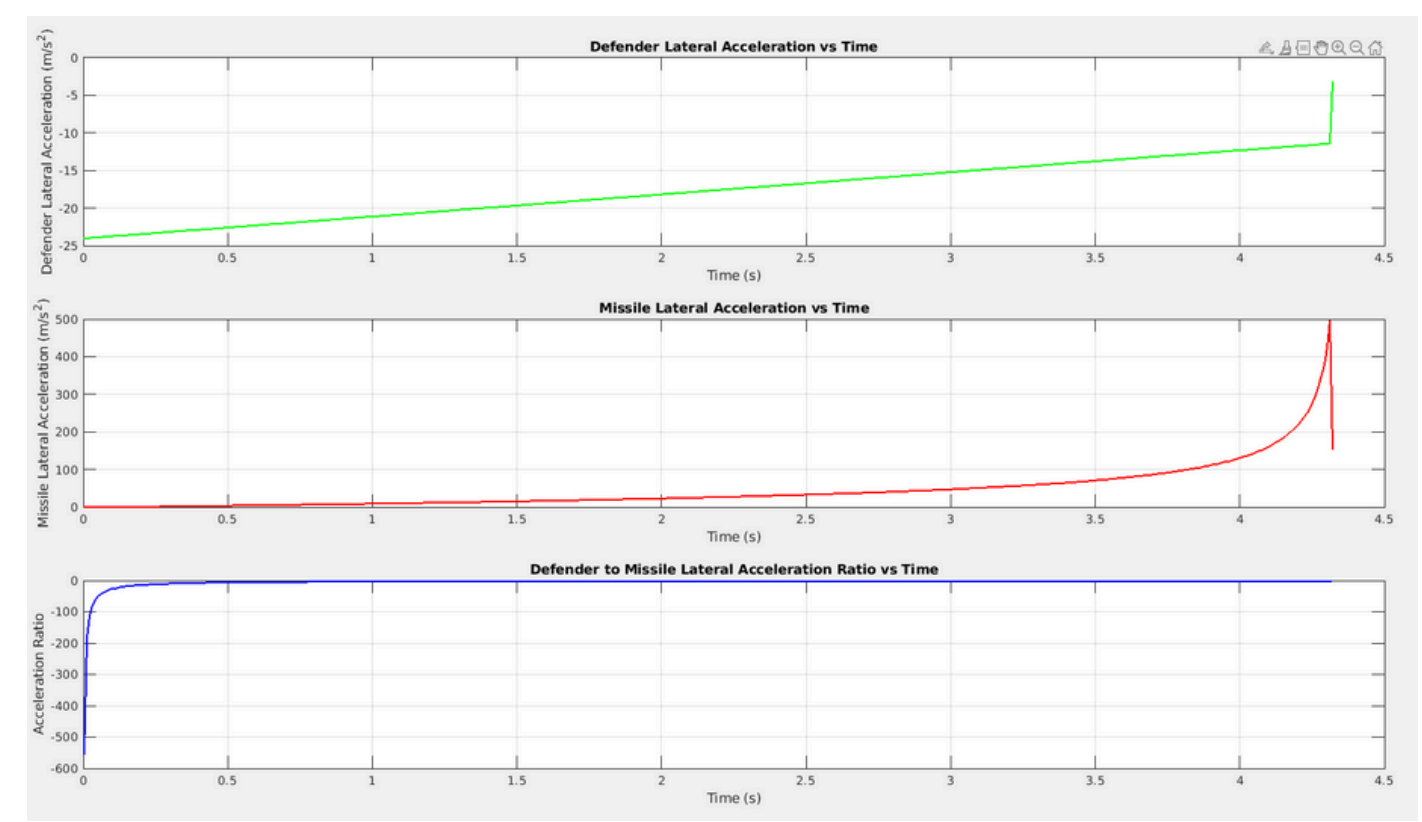
Fig. 6 Side-on attack geometry: sample engagement ($\gamma_d = 19.47$ deg, and $\gamma_m = 180$ deg).



Acceleration Plots given in Paper

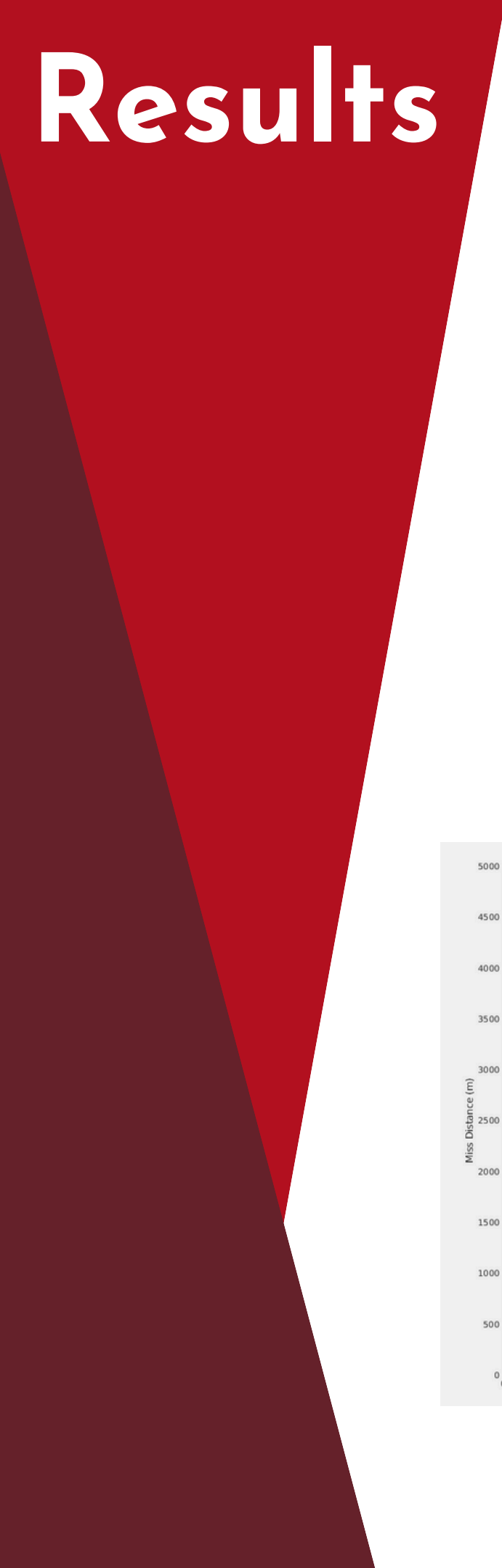


Acceleration Plots with CLOS

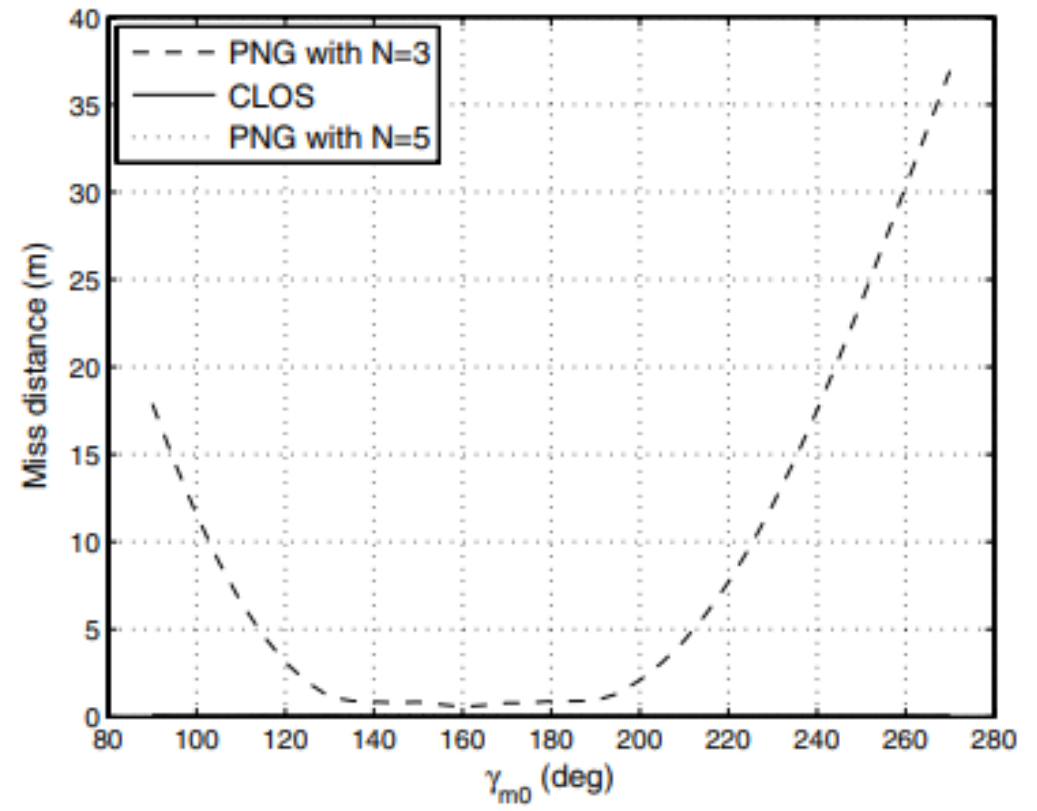


Acceleration Plots with LOS

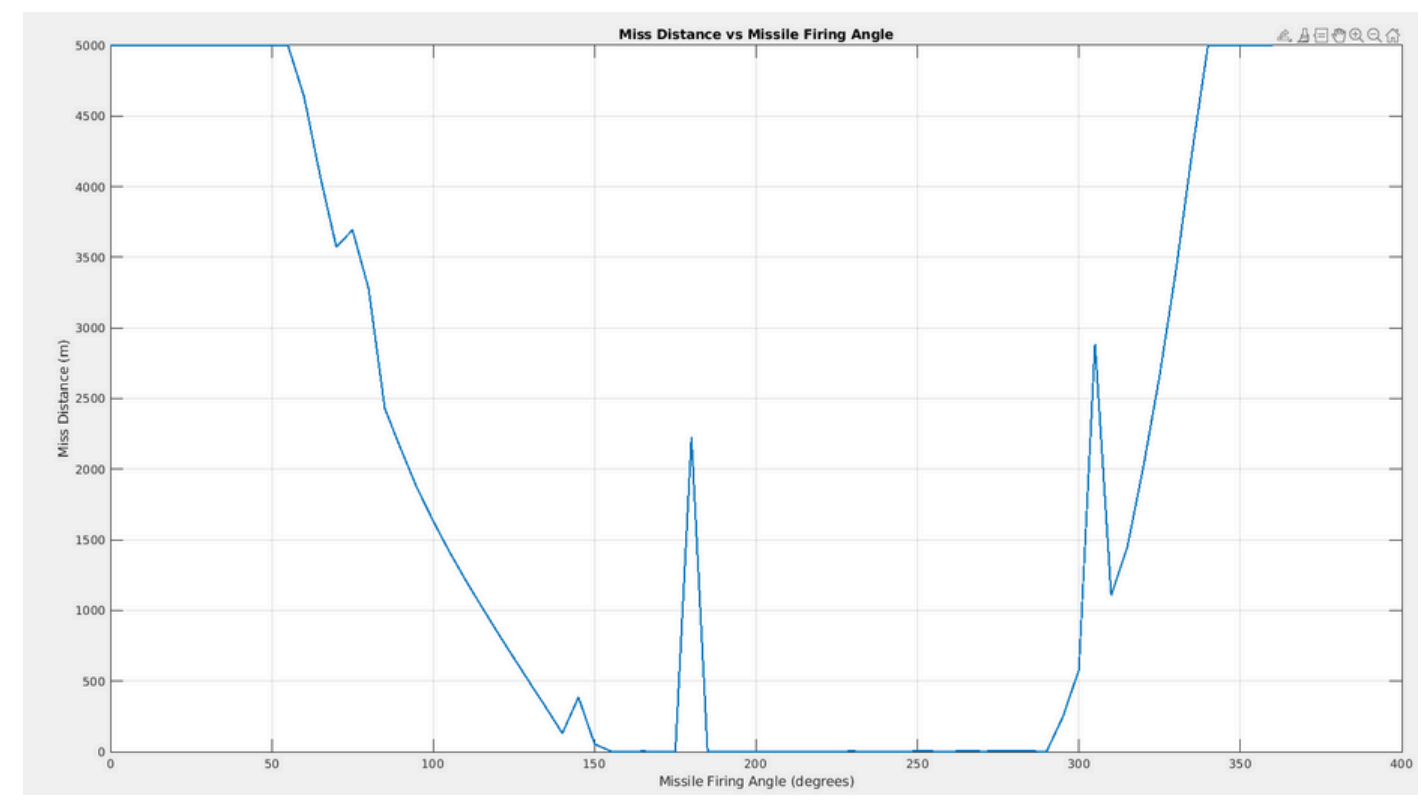
Acceleration Plots with PN



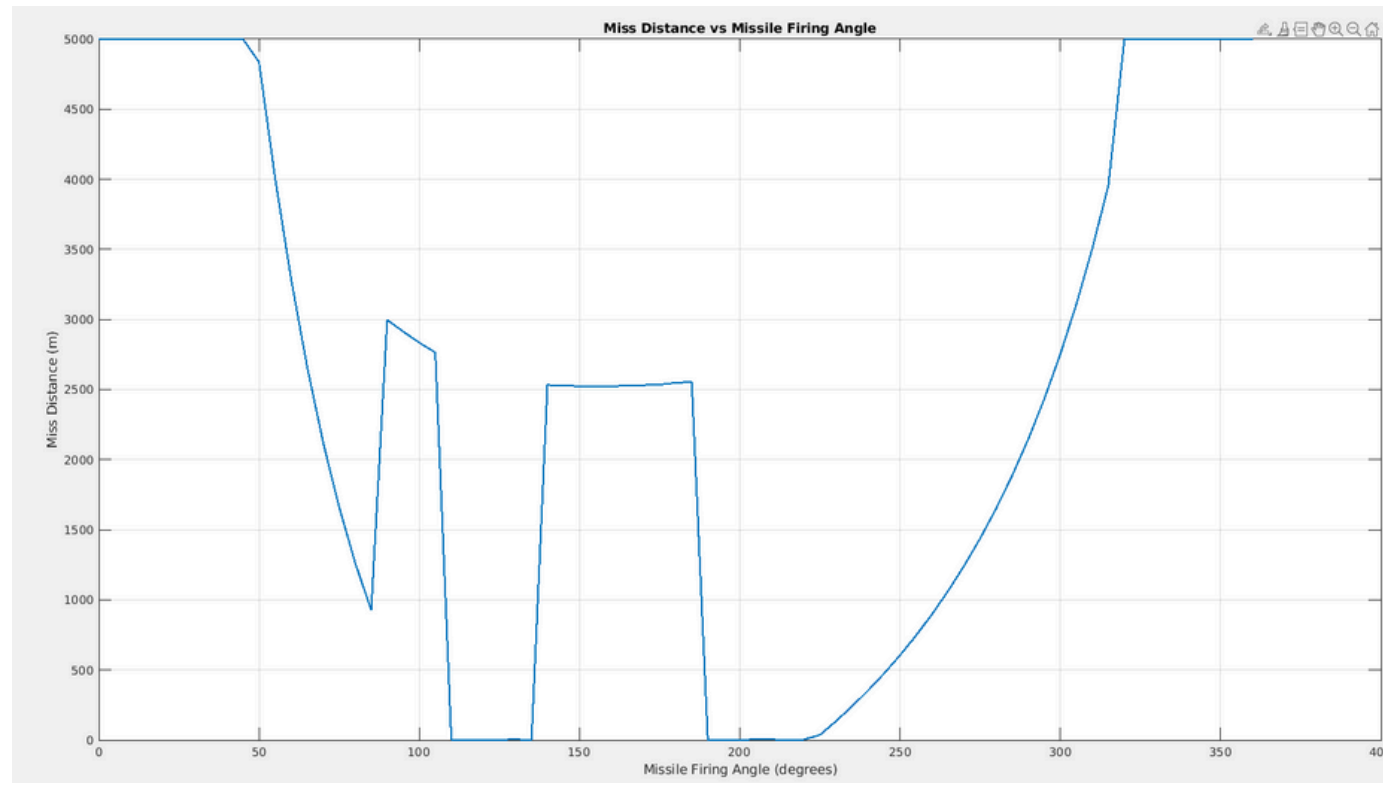
Side-on attack geometry



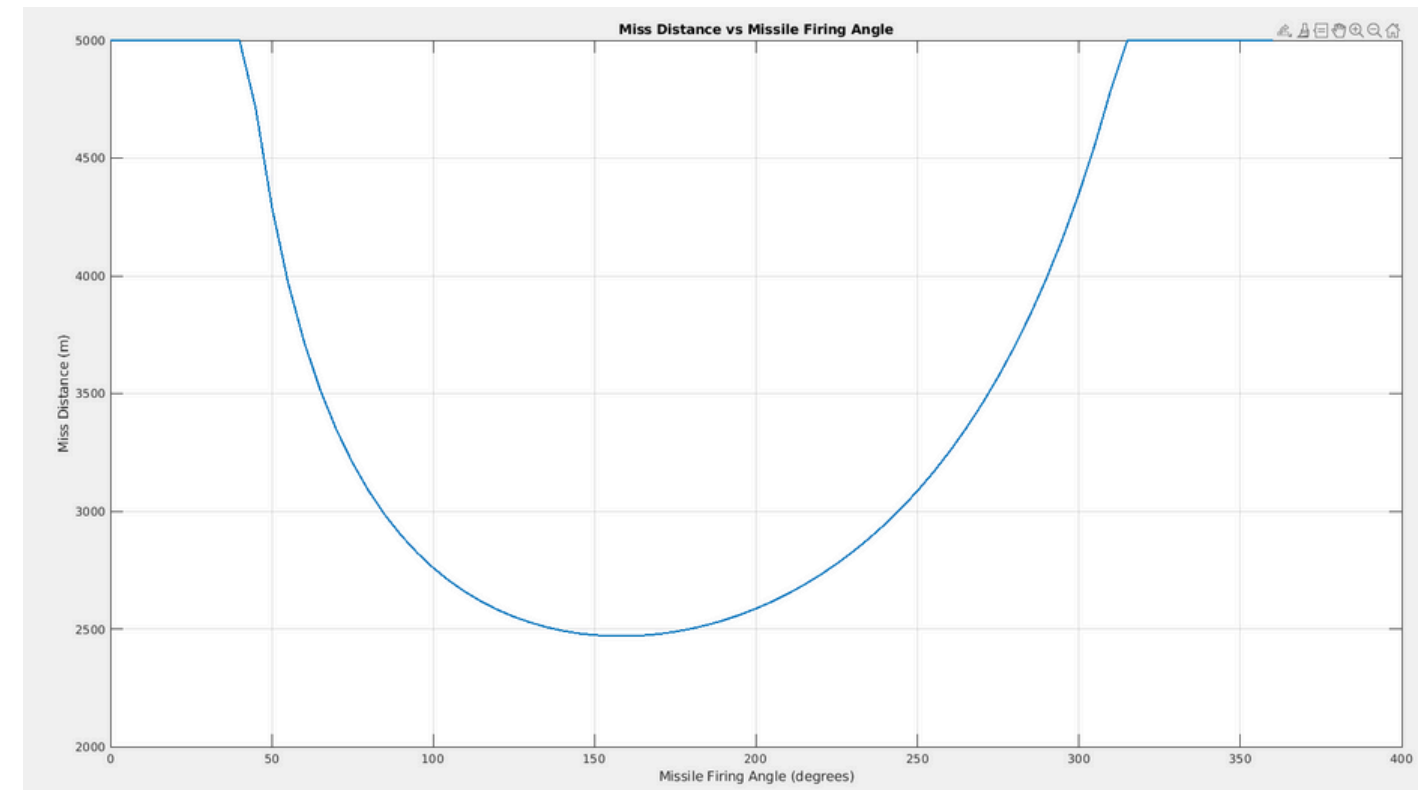
a) Miss distance vs missile firing angle



Miss Distance vs Firing angle in CLOS



Miss Distance vs Firing angle in LOS



Miss Distance vs Firing angle in PN

Further Improvements

- Missile can update its acceleration such that the defender acceleration is dramatically increased putting a load on the defender actuators
- Similar to the derivation of cooperative guidance law, the coefficient of missile lateral acceleration can be found for which ratio is maximum for a non cooperative target.

$$n = \frac{2 \left(\dot{R}_d - \frac{R_d}{R} \dot{R} \right)}{\frac{R_d}{R} - 1}$$

Thank you!